ACADEMIC PROJECT REPORT

France Energy Atlas | 2020–2026

Exploring regional electricity consumption and generation through interactive data storytelling

1. Background, problem and objectives

Electricity is essential to daily life, industry and digital infrastructure, yet demand and generation vary by season, time of day, location and technology. National averages can conceal important regional differences. Open electricity measurements offer a valuable basis for investigation, provided that their temporal resolution, units and completeness are handled correctly.

Research question. How have electricity consumption and generation patterns evolved across metropolitan French regions from 2020 to 2026, and what geographical and temporal differences deserve closer investigation?

The project aims to construct a reproducible dataset, compare regional demand, examine the generation mix, detect recurring temporal patterns and high-value observations, and communicate defensible findings with clear data-quality warnings. The result is an explanatory, descriptive dashboard not a forecasting or causal model.

2. Data provenance and analytical scope

The underlying measurements come from RTE / ODRÉ regional éCO2mix open data: the historical/definitive dataset (eco2mix-regional-cons-def) and the provisional dataset (eco2mix-regional-tr). The project covers 12 metropolitan French regions. The 2026 series is incomplete, so full-year comparisons with earlier years require aligned observation windows.

Validated preparation snapshot	Result
Period	2020–2026 (2026 partial)
Prepared observations	1,463,328
Region-day aggregates	29,856
Missing consumption measurements	108
Reporting regions	12
Quality tests	3 unit tests passed; prepared-data validation passed

3. Data Engineering: from official API to dashboard

Historical processing followed: RTE/ODRÉ API → Python extraction and cleaning → SQLite consolidation → seven-sheet Excel export. Current dashboard preparation follows: existing Excel workbook → Pandas validation and transformation → yearly Parquet files and daily aggregates → Streamlit/Plotly.

Key controls include required region/timestamp identifiers, deduplication by source–region–UTC time, preference for definitive records in overlapping time intervals, treatment of invalid or missing values, and timezone conversion (UTC to Europe/Paris). Power is converted into energy using E (MWh) = P (MW) × interval duration (hours); 15- and 30-minute observations must not be summed as if they represented equal durations. Missing demand is kept missing, not replaced with zero.

4. Narrative design: fourteen connected chapters

The interface follows Hook → Context → Analysis → Insights → Implications. Filters let readers change the region, period and applicable measures; conclusions are recalculated rather than fixed in advance. The sequence deliberately advances from broad context to targeted exploration and transparent interpretation.

Chapter	Purpose and contribution
01 Introduction	Defines the energy question, audience and RTE/ODRÉ provenance.
02 Overview	Time-series line chart of demand energy; identifies selected-period peaks.
03 Historical Evolution	Annual summary table; distinguishes complete from partial years.
04 Regional Ranking	Horizontal bars compare average reported regional demand.
05 Regional Map	Interactive markers locate regions and provide spatial context.
06 Generation Mix	Stacked area chart compares nuclear, wind, solar, hydro, thermal and bioenergy.
07 Hourly Patterns	Weekday × hour heatmap reveals recurring demand rhythms.
08 Seasonal Variability	Monthly box plots show medians, dispersion and outliers.
09 Correlations	Scatter plot and Pearson r explore demand–generation associations.
10 Distribution & Peaks	Histogram and 99th percentile contextualize high readings.
11 Data Quality & EDA	Displays coverage, missingness, duplicate checks and caveats.
12 Conclusions & Actions	Synthesizes selection-dependent findings and next questions.
13 Methodology	Documents extraction, transformations, assumptions and limitations.
14 Student & Credits	Attributes academic work and links to professional profiles.

5. Analytical interpretation and findings

The application combines eight principal chart types: line, bar, map, stacked area, heatmap, box plot, scatter plot and histogram. These complementary views highlight temporal evolution, regional variation, production composition, weekly and seasonal rhythms, linear associations and measurement distributions.

The verified project-level findings concern data readiness and coverage: 1.46 million prepared observations, 12 regions and a transparent incomplete-year warning. Regional leaders, peak dates, renewable shares and correlation coefficients depend on the user's filters; they are intentionally calculated in the app rather than invented as universal results. Regional electricity generation is not necessarily consumed in the same region, and a statistical correlation does not imply causation.

QUALITY, DELIVERY & PERSPECTIVES

6. Data quality, validation and engineering

The preprocessing generated compressed Parquet files and a quality summary. Checks cover observation-key uniqueness, valid 0.25/0.5-hour durations, invalid negative demand, overlap handling and the 108 missing consumption measurements. Automated tests passed locally, and the GitHub Actions quality-check workflow was shown passing on the main branch. These checks establish engineering discipline without guaranteeing that every possible source-data issue has been detected.

The code is organized into app.py, sections/, utils/, scripts/, pipeline/, tests/, docs/ and assets/. The original API-to-SQLite/Excel scripts are preserved under archive/data_engineering/ as evidence of the full historical workflow. Streamlit caches prepared analytical files for responsive interaction.

7. Deployment and planned automation

The application has been deployed at https://france-energy-atlas-fk.streamlit.app/. GitHub Actions currently performs automated tests and dataset-integrity checks. A separate refresh workflow is planned: obtain an actually updated Excel source, rebuild Parquet, validate the files, publish the changes and allow the deployed Streamlit app to load the updated snapshot. Successful CI alone does not prove automatic ingestion of new RTE data; that complete refresh chain remains to be implemented and tested.

8. Limitations and implications

The study is descriptive. Comparisons are affected by incomplete 2026 coverage, occasional missing measurements, mixed source resolutions, regional population and economic differences, and absent weather adjustment. Map markers use representative cities, not exact measurement sites. Chart sampling may be used for performance, while key reported statistical measures use eligible data. The project can support further investigation but should not be used as a standalone grid-operation or energy-policy evaluation tool.

Future extensions could include population-normalized indicators, meteorological context, explicit exchange-flow analysis and a verified incremental data-refresh pipeline. The immediate academic value is a transparent example of connecting public-data engineering with interactive evidence-based storytelling.

9. Alignment with the course rubric

Assessment criterion	Evidence in this project
Data Storytelling (25)	Research question; 14-chapter narrative; dynamic implications
Data Cleaning & Analysis (25)	Official extraction, cleaning, SQLite, Excel, Parquet and validation
Visualization & UX (25)	Eight visual families, map, controls, KPI and explanatory context
Engineering Quality (15)	Modular code, caching, tests and successful CI workflow
Communication (10)	README, detailed docs, storyboard, this report and demo script